AASRA: 6-Model Modular Machine Learning System

Comprehensive Linear Feature Engineering & Exact API Input Source Specification

Project Context: AASRA — Syngenta Biologicals Yield Protection Overwatch | **Team 02:** Divyansh, Rishabh, Sameer, Ishaan, Ritvik
Target Platforms: Google Vertex AI Model Registry, Next.js Live Production Gateway, WhatsApp Conversational Layer
Specification Scope: Exact API endpoints, HTTP methods, request parameters, JSON response keys, and inter-model data contracts.

Architectural Foundation: AASRA decouples agricultural intelligence into 6 specialized, leak-proof models executed in a strict linear pipeline. Every single model is mathematically tuned for its distinct agronomic purpose: Model 1 classifies impending climate stress; Model 2 predicts stomatal readiness; Model 3 ranks biological interventions; Model 4 predicts non-linear dosage response; Model 5 forecasts counterfactual baseline yield; and Model 6 calculates unbiased Return on Biological Investment (ROBI) via Double Machine Learning. Below is the comprehensive directory of exact API input endpoints and the linear model specifications.

1. Master API & Data Telemetry Provenance Directory

All real-time features ingest directly from live, production-grade endpoints with authenticated credentials:

Category & Data Source	Exact Live Endpoints & Protocols	Telemetry Variables Extracted	Downstream Pipeline Consumer
Meteoblue Dataset API **synJg7GEMeblkyn6QY**	POST https://my.meteoblue.com/dataset/query Auth: ?apikey=KEY Fallback: GET /packages/basic-day	Max Temp (Code 11, max), Min Temp (Code 11, min), Mean Temp (Code 11, mean), Rain (Code 61, sum), Soil Moisture 0-10cm (Code 144, mean), ET0 (Code 261), Wind Speed (Code 32).	Feeds Model 1 (Stress Risk), Model 2 (Readiness Gate), Model 5 (Seasonal Baseline).
Syngenta CE Hub API **b5428df1...cb98**	GET https://services.cehub.syngenta-ais.com/api/AgronomicsDecisionRecommendation/ Headers: ApiKey: KEY	/GDDRecommendation: accumulated GDD. /HydricStressRecommendation: water deficit. /SprayWindowRecommendation: Delta-T & spray hours.	Cross-validates Meteoblue telemetry. Dual-sensor consensus triggers high-confidence gates.
Open-Meteo API **Free / Open-Source**	GET https://api.open-meteo.com/v1/forecast Params: hourly=temperature_2m,relative_humidity_2m...	Granular hourly 2m temperature, relative humidity %, precipitation probability %, and surface wind speed (km/h).	Backup & high-resolution time-of-day spray window calculation in Model 2.
ISRIC SoilGrids REST API **Global 250m Resolution**	GET https://rest.isric.org/soilgrids/v2.0/properties/query Properties: phh2o, soc, clay, silt, sand	Soil pH in H2O (0-15cm, /10 scale), Soil Organic Carbon (SOC dg/kg to %), Clay % fraction, Silt %, Sand %.	Feeds Model 3 (soil pH compatibility), Model 4 (SOC nutrient uptake), Model 5 (water retention).
Government Open Data **Agmarknet & data.gov.in**	GET https://api.data.gov.in/resource/... Scraper: https://agmarknet.gov.in/SearchCmmMkt.aspx	Daily APMC mandi modal price (INR/quintal), 5-year seasonal price trends, 10-year district crop yields.	Feeds Model 5 (district counterfactual baseline) and Model 6 (ROBI monetization).
ICRISAT VDSA Database **Open Research Data**	http://data.icrisat.org/dld/ Meso-level district agricultural time series	Rainfed crop yield distributions, input expenditure distributions, extreme dry-spell yield impacts.	Training baseline for Model 5 and confounding covariate weights for Model 6.
Farmer Field Twin **Direct Farmer Inputs**	AASRA Onboarding Form & Mobile Geolocation Local device GPS: navigator.geolocation	Crop variety (Soybean JS-335, Potato, Wheat), Sowing Date, Exact GPS coordinates, Soil Type, Farm Size, Irrigation.	Initiates GDD accumulation clock, anchors GIS queries, and establishes treatment cohorts.

2. Technical API Connection Blueprints & Response JSON Mapping

Below are the verified, exact request contracts, HTTP methods, authentication schemes, and JSON response paths used by AASRA's data ingestion layer to extract input features for model inference:

API Provider	Request Method, URL & Parameters	Request Body / Payload Sample	Exact JSON Response Path Extracted
Meteoblue Dataset API **Weather & Soil Telemetry**	**POST** https://my.meteoblue.com/dataset/query?apikey=synJg7GEMeblkyn6QY Headers: Content-Type: application/json **Critical:** Coordinates are [lon, lat]!	`{ "geometry": {"type": "MultiPoint", "coordinates": [[76.56, 18.40]]}, "timeIntervals": ["2026-06-15T+00:00/2026-06-22T+00:00"], "queries": [{ "domain": "NEMSGLOBAL", "timeResolution": "daily", "codes": [{"code": 11, "level": "2 m above gnd", "aggregation": "max"}, {"code": 11, "level": "2 m above gnd", "aggregation": "min"}, {"code": 61, "level": "sfc", "aggregation": "sum"}, {"code": 144, "level": "0-10 cm down", "aggregation": "mean"}] }] }`	• temp_max: data[0].coordinates[0].dates[i].value (Code 11 max) • temp_min: data[1].coordinates[0].dates[i].value (Code 11 min) • rainfall: data[2].coordinates[0].dates[i].value (Code 61 sum) • soil_moisture: data[3].coordinates[0].dates[i].value (Code 144 mean)
Syngenta CE Hub API **GDD & Phenology**	**GET** /api/AgronomicsDecisionRecommendation/GDDRecommendation Headers: ApiKey: b5428df1-abb7-4f52-8a13-ddaed67dcb98 Query: latitude, longitude, startDate, endDate, baseLimit=10, maxLimit=35, useEnhancedFormula=true	*No body (HTTP GET query parameters).* ***Constraint:*** *Range must be entirely in past or entirely in future.*	• crop_gdd_accumulated: response[last].accumlatedValue • daily_gdd_rate: response[i].value
Syngenta CE Hub API **Spray Window Gate**	**GET** /api/AgronomicsDecisionRecommendation/SprayWindowRecommendation Headers: ApiKey: b5428df1...cb98 Query: latitude, longitude, startDate, endDate, sprayingType=Biological, top=5, format=json	*No body (HTTP GET query parameters).*	• spray_window_safe: response[0].isFavorable • delta_t_celsius: response[0].deltaT • optimal_spray_hour: response[0].bestApplicationTime
Syngenta CE Hub API **Hydric Drought Stress**	**GET** /api/AgronomicsDecisionRecommendation/HydricStressRecommendation Headers: ApiKey: b5428df1...cb98 Query: latitude, longitude, startDate, endDate, waterAvailabilty=50	*Note official API query parameter spelling: waterAvailabilty (missing 'i').*	• hydric_stress_level: response[0].stressIndex • drought_deficit_pct: response[0].waterDeficit
ISRIC SoilGrids REST API **Chemical & Physical Soil**	**GET** https://rest.isric.org/soilgrids/v2.0/properties/query Query: lat=18.40&lon;=76.56&property;=phh2o&property;=soc&property;=clay&depth;=0-5cm&depth;=5-15cm&value;=mean	*No body (Public REST API, no auth required).*	• soil_ph: layers[name='phh2o'].depths[0].values.mean / 10.0 • soil_organic_carbon_pct: layers[name='soc'].depths[0].values.mean / 100.0 • soil_clay_pct: layers[name='clay'].depths[0].values.mean / 10.0
Agmarknet / data.gov.in **Mandi Commodity Pricing**	**GET** https://api.data.gov.in/resource/9ef84268-d588-465a-a308-a864a43d0070 Query: api-key=KEY&format;=json&filters;[state]=Maharashtra&filters;[commodity]=Soyabean	*No body (Government API key passed via query string).*	• mandi_price_per_q: records[0].modal_price (INR/Quintal) • mandi_min_price: records[0].min_price • mandi_max_price: records[0].max_price

3. Linear Model Specifications — Stage 1 & Stage 2 (PS-02 Early Warning & Readiness)

Stage 1: Model 1 — Climate Stress Early Warning Classifier (PS-02)

Model Function: Predicts imminent abiotic stress 3 to 7 days before visual damage occurs. **Algorithm:** XGBClassifier(n_estimators=200, max_depth=5, learning_rate=0.05) (Classes: 0=Optimal, 1=Heat, 2=Drought, 3=Compound). **Assigned Owner:** Divyansh.

Feature Name	Type & Range	Exact API Endpoint & Query	Exact JSON Response Path / Extraction Logic
temp_max_forecast_7d	Float (°C) 20.0 to 48.0	**Meteoblue Dataset API:** POST /dataset/query Domain: NEMSGLOBAL, Code 11, max	response['data'][0]['coordinates'][0]['dates'][i]['value'] Aggregated: max(values[:7]) across 7-day forecast horizon.
temp_night_min_7d	Float (°C) 12.0 to 32.0	**Meteoblue Dataset API:** POST /dataset/query Domain: NEMSGLOBAL, Code 11, min	response['data'][1]['coordinates'][0]['dates'][i]['value'] Aggregated: mean(values[:7]). Night temp > 24.5°C blocks cellular recovery.
rh_avg_forecast_7d	Float (%) 15.0 to 98.0	**Open-Meteo Hourly API:** GET /v1/forecast?hourly=relative_humidity_2m Fallback: Meteoblue Packages API	response['hourly']['relative_humidity_2m'][:168] Extracted: mean(hourly_rh) over 7-day window.
vpd_kpa	Float (kPa) 0.5 to 5.5	**Derived via Tetens Physics:** Calculated from Meteoblue temp_max and Open-Meteo rh_avg	svp = 0.61078 * exp((17.27 * T) / (T + 237.3)) vpd = svp * (1.0 - rh / 100.0). Primary biophysical driver of transpiration shutdown.
soil_moisture_vol_pct	Float (%) 8.0 to 50.0	**Meteoblue Dataset API:** Code 144 (0-10 cm down, mean) Cross-checked: **CE Hub API** HydricStress	response['data'][3]['coordinates'][0]['dates'][i]['value'] Cross-validated with CE Hub: 100 - response[0]['waterDeficit']. < 20% indicates acute drought.
consecutive_hot_days	Integer (Days) 0 to 14	**Derived Rolling Time-Series:** Meteoblue daily temperature history + forecast	Count of consecutive days where daily_tmax > 35.0°C. Prolonged continuous heat causes cellular protein coagulation.
crop_gdd_accumulated	Float (°C-days) 0 to 2500	**Syngenta CE Hub API:** GET /GDDRecommendation Params: startDate={sowing_date}, baseLimit=10	response[-1]['accumlatedValue'] Sowing date starts clock. Accurately maps crop phenology stage.

Model 1 Output Handed Downstream: stress_type (1=Heat, 2=Drought, 3=Compound) | stress_intensity_index (0.0 to 1.0 probability) | days_to_impact (3 to 7 days).

Stage 2: Model 2 — Biological Intervention Readiness Engine (PS-02)

Model Function: Evaluates whether the field microclimate permits safe, effective foliar biostimulant uptake. **Algorithm:** CalibratedClassifierCV(LogisticRegression, method='sigmoid') (Platt Scaling) with biophysical threshold cutoffs. **Assigned Owner:** Rishabh.

Feature Name	Type & Gate Rule	Exact API Endpoint & Query	Exact JSON Response Path / Extraction Logic
soil_moisture_pct	Float (%) Optimal: 40% - 70%	**Meteoblue API:** Code 144 (0-10cm) & **CE Hub API:** /HydricStressRecommendation	Meteoblue: response['data'][3]['coordinates'][0]['dates'][0]['value'] If soil < 30%, root xylem pressure collapses; stomata close and foliar sprays fail.
delta_t_celsius	Float (°C) Safe: 2.0°C to 8.0°C	**Syngenta CE Hub API:** GET /SprayWindowRecommendation Fallback: Derived via Stull formula from Meteoblue	CE Hub: response[0]['deltaT'] Atmospheric wet-bulb depression. Delta-T > 8°C causes spray droplet evaporation before leaf penetration. < 2°C causes runoff.
wind_speed_kmh	Float (km/h) Safe: < 15.0 km/h	**Meteoblue Dataset API:** Code 32 (10m) & **Open-Meteo Hourly API**	response['hourly']['wind_speed_10m'][0] High wind causes microscopic droplet drift into non-target zones. Wind > 18 km/h forces readiness to 0.0.
rain_prob_next_48h	Float (%) Safe: < 40%	**Open-Meteo Hourly API:** GET /v1/forecast?hourly=precipitation_probability	max(response['hourly']['precipitation_probability'][:48]) Rainfastness gate: biological peptides require 2-4 hours rain-free to absorb onto foliage.
crop_stage_sensitivity	Float Multiplier 0.2 (Veg) to 1.0 (Flower)	**Derived Phenology Lookup:** Mapped from CE Hub accumlatedValue GDD	STAGE_SENSITIVITY[current_stage] (Vegetative: 0.2, Flowering: 1.0, Pod fill: 0.85). Focuses investment where economic ROI is maximized.

Model 2 Output Handed Downstream: readiness_score (0.0 to 1.0 calibrated probability) | spray_window_safe (True/False) | delta_t (current wet-bulb depression).

4. Linear Model Specifications — Stage 3 & Stage 4 (PS-03 Portfolio & Response Curve)

Stage 3: Model 3 — Biological Product Ranking Engine (PS-03)

Model Function: Ranks Syngenta's catalog of 50 biological products to select the optimal Top-3 recommendations based on stress type, crop stage, and soil chemistry. **Algorithm:** XGBRanker(objective='rank:ndcg', eval_metric='ndcg@3') (LambdaMART pairwise ranking). **Assigned Owner:** Sameer (Team Leader).

Feature Name	Type & Range	Interconnected Source / Provenance	Exact Ingestion Path / Role in Pairwise Ranker
stress_type	Integer Category [1, 2, 3]	**UPSTREAM OUTPUT: MODEL 1** (Divyansh's Stress Classifier)	model1_output['stress_class'] Restricts ranking pool to products formulated for the active stress (e.g., Quantis for Heat, Isabion for Drought/Osmotic).
readiness_score	Float 0.0 to 1.0	**UPSTREAM OUTPUT: MODEL 2** (Rishabh's Readiness Engine)	model2_output['readiness_score'] Down-weights foliar biostimulants if stomata are closed; promotes soil-drench or systemic alternatives.
crop_stage	Categorical Enum Germ, Veg, Flower, Pod	**Derived Phenology Engine:** CE Hub API /GDDRecommendation	gdd = cehub_response[-1]['accumlatedValue'] Mapped to crop stage table (e.g. Soybean JS-335: Flower = 450-700 GDD). Checks product label stage compatibility.
soil_ph	Float 5.5 to 8.5	**ISRIC SoilGrids REST API:** GET /properties/query?property=phh2o (or Farmer Field Twin Input)	response['properties']['layers'][0]['depths'][0]['values']['mean'] / 10.0 Alkaline soil (pH > 7.5) degrades soil-applied peptides; forces ranker to elevate foliar formulations.
product_catalog_vector	Vector (50 Items) Active salts, cost/acre	**Syngenta Product Knowledge Base:** Verified catalog from syngenta.co.in	Encoded feature vector for each of the 50 products: amino acid profile, free peptide chain length, potassium/calcium salts, dealer MRP.

Model 3 Output Handed Downstream: selected_product_key ('quantis') | top_ranked_products (1st Quantis 94.2%, 2nd Isabion 88.5%, 3rd Amistar Top 71.0%) | active_salts ('Amino acids + K + Ca').

Stage 4: Model 4 — Expected Product Response & Yield Protection Curve (PS-03 / PS-07)

Model Function: Predicts the non-linear yield recovery curve (+% yield protected or +quintals/acre saved) based on dosage, timing, and soil carbon. **Algorithm:** CatBoostRegressor(iterations=250, loss_function='Huber:delta=1.5') with Ordered Target Encoding. **Assigned Owner:** Sameer (Team Leader).

Feature Name	Type & Range	Interconnected Source / Provenance	Exact Ingestion Path / Law of Diminishing Returns
product_key	Categorical String e.g. 'quantis', 'isabion'	**UPSTREAM OUTPUT: MODEL 3** (Top-1 recommended product)	model3_output['selected_product_key'] Passed as native categorical string to CatBoost. Identifies physiological mechanism without one-hot leakage.
dosage_ml_per_acre	Float 200 to 600 ml/acre	**Farmer Interactive Slider:** Live Web Frontend Input	frontend_request['dosage_ml_per_acre'] Models Mitscherlich's Law of Diminishing Marginal Returns: increments past 400ml produce progressively smaller yield gains.
timing_days_before_stress	Integer (Days) 1 to 5 days prior	**UPSTREAM OUTPUT: MODEL 1** (`days_to_impact` from weather forecast)	model1_output['days_to_impact'] Proactive preventative spraying (48h before heatwave) provides 3x higher cellular protection than reactive curative spraying.
stress_intensity_index	Float 0.0 to 1.0	**UPSTREAM OUTPUT: MODEL 1** (Model 1 prediction confidence)	model1_output['confidence'] Quantifies severity of impending stress. Higher stress provides higher potential headroom for yield protection.
soil_organic_carbon_pct	Float (%) 0.2% to 1.8%	**ISRIC SoilGrids REST API:** GET /properties/query?property=soc	response['properties']['layers'][0]['depths'][0]['values']['mean'] / 100.0 Soil microbial vitality: higher SOC increases foliar product translocation efficiency.

Model 4 Output Handed Downstream: delta_yield_pct (+14.5% yield loss prevented) | optimal_dosage_ml (400 ml/acre) | marginal_gain_curve ([0.5L: +8.0%, 1.0L: +14.5%, 1.5L: +16.2%]).

5. Linear Model Specifications — Stage 5 & Stage 6 (PS-07 Baseline & Causal ROBI)

Stage 5: Model 5 — Field Yield Baseline Prediction (PS-07)

Model Function: Predicts the unperturbed counterfactual yield: 'What would this farm produce normally under this season\'s weather without biologicals?' **Algorithm:** XGBRegressor(n_estimators=300, max_depth=6, learning_rate=0.04) (GroupKFold by District). **Assigned Owner:** Ishaan.

Feature Name	Type & Range	Exact API Endpoint & Query	Exact JSON Response Path / Ingestion Logic
gdd_seasonal_total	Float (°C-days) 1200 to 2400	**Meteoblue ERA5 Dataset API:** POST /dataset/query Domain: ERA5 (Season-to-date daily)	Cumulative sum of max(0, (tmax + tmin)/2 - 10) across entire season. Measures total thermal potential.
rainfall_total_mm	Float (mm) 200 to 1400	**Meteoblue Dataset API:** Code 61 (sfc, sum) Cross-checked: CE Hub API	sum(response['data'][2]['coordinates'][0]['dates'][i]['value']) Seasonal cumulative precipitation. Baseline rainfed yield determinant.
dry_spell_max_consecutive_days	Integer (Days) 0 to 35	**Derived Time-Series Metric:** Meteoblue precipitation time-series	Longest continuous run of days where daily rainfall < 2.5mm during critical growth phases. Primary driver of catastrophic yield loss.
extreme_heat_days_count	Integer (Days) 0 to 20	**Derived Time-Series Metric:** Meteoblue temperature time-series	Count of days where tmax > 38.0°C during flowering/anthesis. Drives floret sterility.
soil_clay_pct	Float (%) 10.0% to 65.0%	**ISRIC SoilGrids REST API:** GET /properties/query?property=clay	response['properties']['layers'][0]['depths'][0]['values']['mean'] / 10.0 Soil water holding capacity: Vertisols (black clay) buffer dry spells; sandy soils dry out immediately.
district_historical_mean_yield	Float (Q/Acre) 3.0 to 45.0	**data.gov.in / ICRISAT VDSA:** Ministry of Agriculture Crop Statistics	data_gov_record['yield_q_per_acre'] 10-year official district historical average. Calibrates local technological and agro-climatic baseline.

Model 5 Output Handed Downstream: expected_baseline_yield_q (e.g. 21.0 Quintals/Acre) | counterfactual_revenue_baseline (Baseline Yield x Mandi Price).

Stage 6: Model 6 — Causal Biological Impact & ROBI Attribution Engine (PS-07)

Model Function: Isolates true causal treatment effect (tau) by partialling out wealth and weather confounders, computing unbiased Return on Biological Investment. **Algorithm:** LinearDML(model_y=RandomForestRegressor, model_t=RandomForestClassifier, cv=3) via EconML. **Assigned Owner:** Ritvik.

DML Covariate Role	Variable Schema	Interconnected & API Source	Exact Ingestion Path / Confounder Partialling
Treatment (T)	treatment_applied Binary (1 or 0)	**Farmer Application Log** or Verified Syngenta Purchase	farmer_profile['treatment_applied'] Core causal binary treatment variable.
Outcome (Y)	observed_yield Float (Quintals/Acre)	**Farmer Harvest Record** (Inference: M5 * (1 + M4 Delta))	farmer_profile['harvest_yield'] or predicted outcome model5_yield * (1 + model4_delta).
Confounders (W) **(Nuisance Covariates)**	• rainfall_total • soil_moisture • irrigation_type • farm_size_acres	**Meteoblue API** (weather) + **Farmer Profile** (wealth)	Cross-fitted via auxiliary ML models. Ensures richer farmers with borewells do not give false credit to the biostimulant.
Heterogeneity (X) **(Effect Modifiers)**	• stress_intensity • crop_growth_stage • soil_type	**OUTPUT FROM MODEL 1** + **Farmer Field Twin**	Estimates Conditional Average Treatment Effect (CATE): proves product delivers higher recovery under extreme stress.
Mandi Market Price	mandi_price_per_q Float (INR/Quintal)	**Agmarknet APMC Mandi API:** data.gov.in daily market prices	response['records'][0]['modal_price'] Live wholesale price. Converts physical yield saved into verified rupees.
Product Cost	product_cost_per_acre Float (INR/Acre)	**Syngenta Dealer MRP:** Catalog Knowledge Base	catalog[product_key]['dealer_mrp'] * dosage Computes net profit: ROBI = (tau * Mandi_Price - Cost) / Cost.

Model 6 Output Handed Downstream: causal_yield_gain_q (+2.8 Q/Acre true causal treatment effect tau) | revenue_saved_inr (INR 11,350 net profit) | robi_multiplier (20.6x Return on Biological Investment).

6. Interconnection Architecture & Unified JSON Contract

The 6 models execute in a tightly coupled mathematical chain. Upstream outputs become mandatory downstream inputs:

Pipeline Transition	Upstream Model Output	Downstream Model Consuming Feature	Agronomic & Mathematical Function
Stage 1 → Stage 3	Model 1: stress_type & stress_intensity	Model 3: stress_type in ranking vector	Filters catalog so ranker only evaluates biostimulants designed to neutralize the active stress.
Stage 2 → Stage 3	Model 2: readiness_score & spray_window_safe	Model 3: readiness_score weight in LambdaMART	Suppresses foliar products if field is closed to spray; boosts systemic alternatives.
Stage 3 → Stage 4	Model 3: selected_product_key ('quantis')	Model 4: product_key categorical feature	Directs CatBoost to evaluate the specific dose-response curve for the #1 chosen product.
Stage 1 → Stage 4	Model 1: days_to_impact (lead time)	Model 4: timing_days_before_stress feature	Allows CatBoost to calculate proactive vs reactive efficacy penalty curves.
Stage 4 + 5 → Stage 6	Model 4: delta_yield_pct Model 5: expected_baseline_yield_q	Model 6: observed_yield / treatment outcome matrix	Provides unperturbed baseline and expected recovery to Double ML to isolate causal tau.
Stage 6 → Gemini	Model 6: tau, revenue_saved_inr, robi_multiplier	**Google Gemini 2.5 Flash** & **WhatsApp Bot**	Gemini translates raw mathematical JSON into warm, trustworthy vernacular farmer dialogue.

Standardized Unified JSON Payload Contract

Upon completing the 6-stage linear pipeline, the system outputs the following unified JSON payload consumed by the frontend and Gemini 2.5:


```
{
  "farmer_id": "MH-LAT-2026-8842",
  "location": { "lat": 18.4088, "lon": 76.5604, "district": "Latur", "state": "Maharashtra" },
  "field_twin": { "crop": "Soybean", "variety": "JS-335", "sowing_date": "2026-06-15", "stage": "R1 Flowering", "soil_ph": 8.1 },
  "stage1_model1_risk": {
    "stress_type": "Compound Heat & Drought", "stress_class": 3, "confidence": 0.92, "days_to_impact": 4, "peak_forecast_temp": 42.4
  },
  "stage2_model2_readiness": {
    "spray_window_safe": true, "readiness_score": 0.88, "delta_t_celsius": 5.2, "optimal_spray_window": "06:30 AM - 09:15 AM Tomorrow"
  },
  "stage3_model3_portfolio": {
    "recommended_products": [
      { "rank": 1, "product_key": "quantis", "name": "Quantis", "score": 94.2, "active_salts": "Amino acids + K + Ca", "cost_per_acre": 550 },
      { "rank": 2, "product_key": "isabion", "name": "Isabion", "score": 88.5, "active_salts": "Animal Origin Peptides", "cost_per_acre": 400 },
      { "rank": 3, "product_key": "amistar_top", "name": "Amistar Top", "score": 71.0, "active_salts": "Azoxystrobin + Difenoconazole", "cost_per_acre": 750 }
    ]
  },
  "stage4_model4_response": {
    "product": "Quantis", "optimal_dosage_ml_per_acre": 400, "yield_protected_pct": 14.5, "timing_advantage": "3.2x higher efficacy than curative"
  },
  "stage5_model5_baseline": {
    "counterfactual_baseline_yield_q_per_acre": 21.0, "district_10yr_norm_q": 19.8
  },
  "stage6_model6_causal_robi": {
    "causal_treatment_effect_tau_q": 2.8, "mandi_price_per_q": 4250, "product_cost_inr": 550, "net_revenue_saved_inr": 11350, "robi_multiplier": "20.6x ROI"
  }
}
```

7. Training vs. Live Web Inference Pipelines & Leakage Prevention

To guarantee high real-world accuracy on the website without target leakage or overfitting, training and live web inference use distinct pipelines:

Model	Training Phase Data & Provenance	Live Web Inference Data & Ingestion	Target Evaluation Metric
Model 1 **Stress Classifier**	10-Year historical ERA5 hourly weather (1985-2025) aligned with IMD historical heatwave & drought records.	Live 7-day weather forecast from Meteoblue Dataset API + CE Hub ShortRangeForecastDaily.	Macro F1 > 0.85 ROC-AUC > 0.88
Model 2 **Readiness Engine**	Historical spray field trials and stomatal conductance logs from ICAR / Syngenta protocol data.	Live 48h hourly microclimate from Meteoblue & Open-Meteo (Delta-T, wind speed, rain probability).	Brier Score < 0.08 LogLoss < 0.25
Model 3 **Product Ranker**	Pairwise multi-location trial outcomes from Syngenta product registration dossiers.	Model 1 stress output + Model 2 readiness output + SoilGrids API pH + Catalog Vector.	NDCG@3 > 0.92
Model 4 **Response Curve**	Syngenta multi-dosage field trials (0.5L, 1.0L, 1.5L, 2.0L) under controlled abiotic stress conditions.	Model 3 top product key + Farmer interactive dosage slider + Model 1 lead time + SoilGrids SOC.	R2 > 0.80 RMSE < 1.5%
Model 5 **Baseline Regressor**	10-year district crop production statistics from data.gov.in & ICRISAT VDSA rainfed time-series.	Season-to-date accumulated GDD & rainfall from Meteoblue ERA5 + SoilGrids clay %.	R2 > 0.78 RMSE < 2.2 Q/Acre
Model 6 **Double ML ROBI**	Observational farm survey dataset with treated and untreated control plots (ICRISAT + Syngenta Club).	Model 5 baseline + Model 4 delta yield + Agmarknet APMC live mandi price + Syngenta dealer MRP.	Unbiased causal tau with 95% Confidence Interval

Quality Assurance & The 5 Anti-Leakage Rules

• **1. GroupKFold by District (Spatial Leakage Shield):** Spatial autocorrelation causes adjacent fields to experience identical weather. Models 1, 3, and 5 are strictly cross-validated by District grouping (GroupKFold(n_splits=5, groups=df['district'])).

• **2. TimeSeriesSplit (Temporal Leakage Shield):** Weather and yield series are never randomly shuffled. Shuffling leaks future weather into past models. Training uses 2015-2024; evaluation uses 2025-2026.

• **3. Scaler Pre-Fit Isolation:** All scalers (StandardScaler, MinMaxScaler) are fit strictly on X_train and applied onto X_test without recomputing means or standard deviations.

• **4. Double ML Cross-Fitting for True Causation:** Richer farmers who buy biostimulants also have drip irrigation. Naive comparison assigns credit for water to the biostimulant. Model 6 uses LinearDML with cross-fitting to partial out confounders orthogonally.

• **5. Golden Anti-Cheat Rule:** Any tabular model claiming > 98% accuracy is flagged for target leakage. Agricultural biological systems are naturally noisy; true uncheated models exhibit realistic F1 > 0.85 and R2 > 0.78-0.82.

Production Sign-Off: This linear specification provides the complete engineering blueprint for data ingestion, model training, and Google Vertex AI containerized deployment. All inputs have verified physical sources, exact API response mapping, and all model boundaries are protected against target leakage.